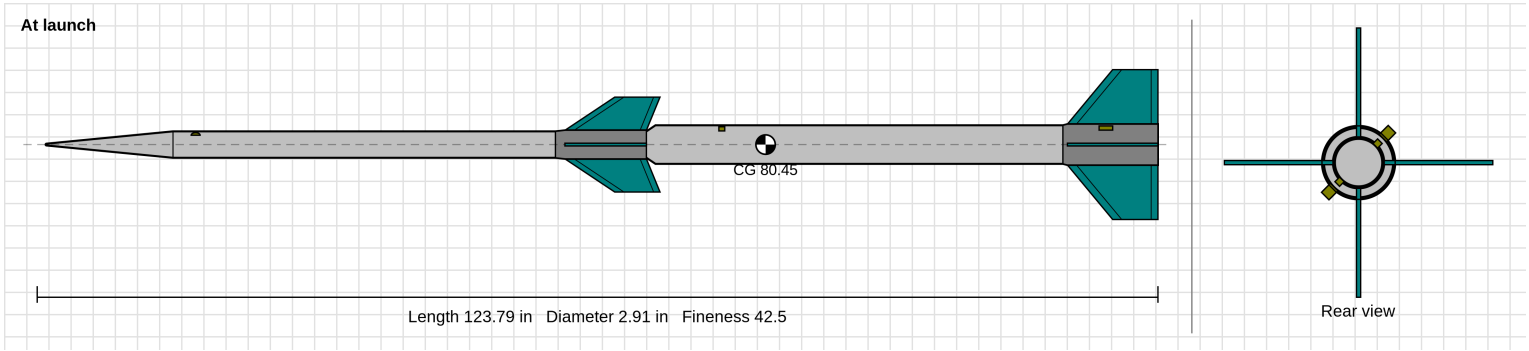
PANDORA - AMA_CONFIG_15 - Results



PANDORA - AMA_CONFIG_15
Nose radius for heating: 0.1 in

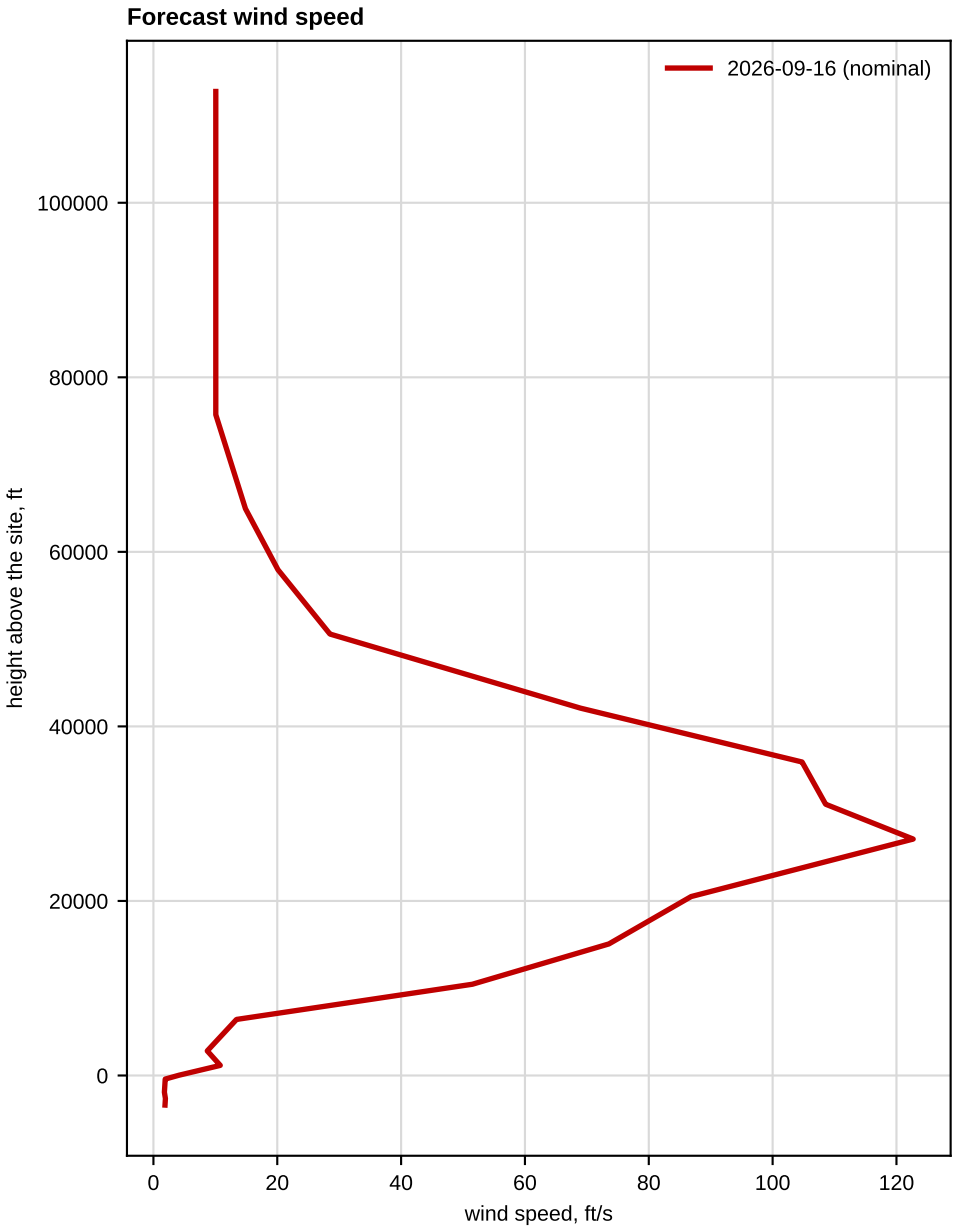
	nominal	
Rail exit speed	115.27	ft/s
Rail exit AoA	2.0079	deg
Forecast gust	4.2651	ft/s
Rail exit AoA in gust	2.1195	deg
Apogee	4.0451e+05	ft
Time to apogee	172.08	s
Max speed	6079.3	ft/s
Max acceleration	106.84	g
Max roll rate	0.0073785	rev/s
Max angle of attack	0.74394	deg
Max AoA after burnout	0.019062	deg
Roll-pitch crossings	0	
Resonance time	nan	s
Resonance altitude	nan	ft
AoA rise at resonance	nan	deg
Roll lock-in held	0	s
Lock-in starts	nan	s
Max Mach	6.2114	
Max dynamic pressure	8892.6	psf
Peak heat flux	952.62	BTU/ft²/s
Heat load	6551	BTU/ft²
Tilt at ignition	5.2058	deg
Altitude at ignition	31356	ft
Flight time	566.37	s
Landing east	2.4268	mi
Landing north	-23.297	mi
Impact speed	35.421	ft/s
Launch mass	71.337	lb, CG 80.45 in from the nose tip
Min static margin	-8.6168	cal (while loaded)

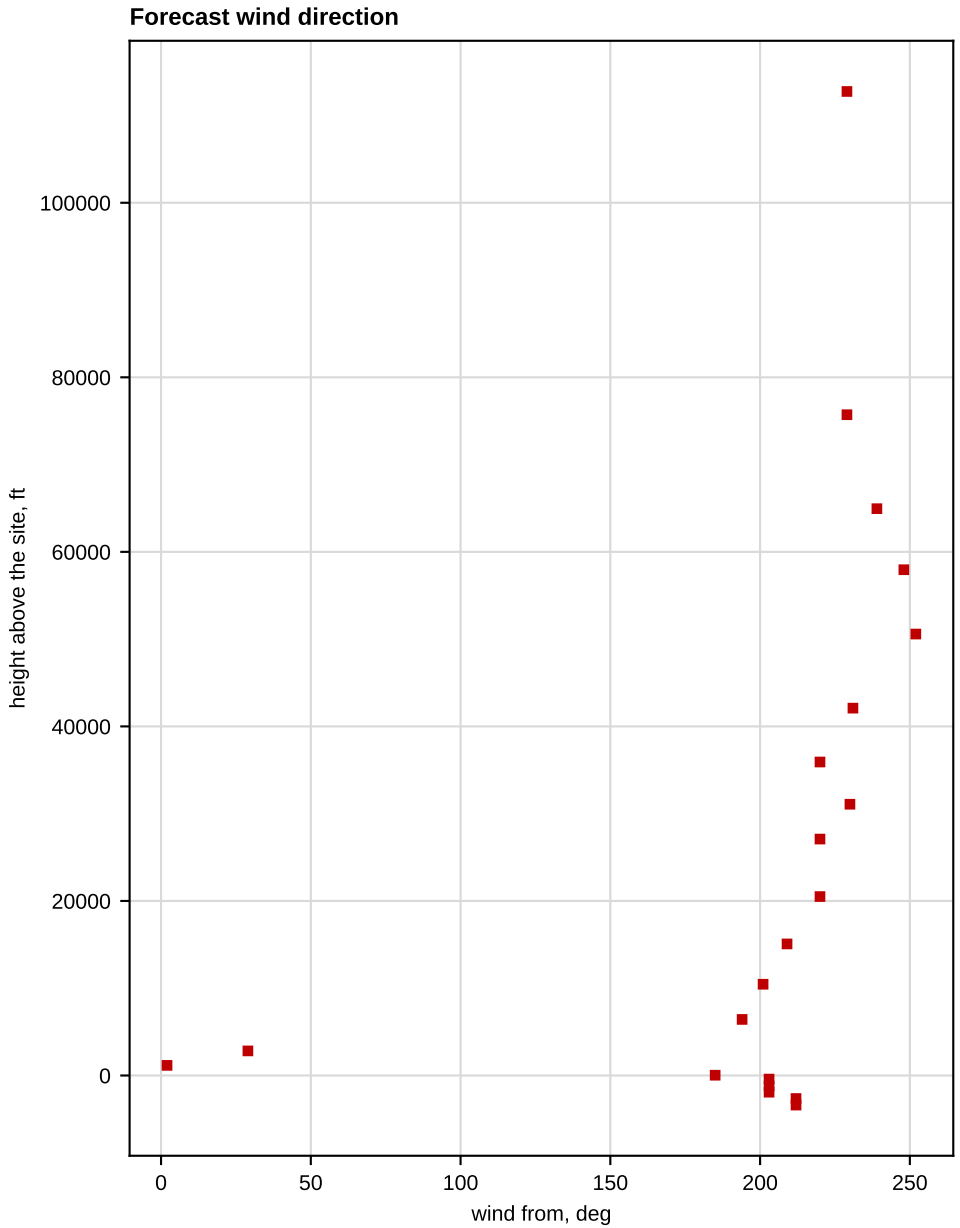
  Note: DATCOM's stack CP jumps 12.9 cal forward between Mach 1.3 and 1.32, where this minimum falls. This is DATCOM's own output (on a stack, from adding the
Angles of attack are while loaded, launch to apogee: dynamic pressure above 5% of its peak.
AoA limit 2 deg: the nominal flight reaches 0.744 deg, under
Sustainer ignited: yes
Launch cylinder, 17 nmi: the nominal flight lands OUTSIDE

Events
      3.43 s Booster burnout, separation
     18.37 s Sustainer ignition

PANDORA - AMA_CONFIG_15 - Weather

```
Weather
  Open-Meteo forecast at 40.8707, -119.1069, 06:00 America/Los_Angeles
  day                   surface wind         gust at the launch hour
  2026-09-16      1.9 ft/s from 212 deg    4.3 ft/s from 185 deg at 06:00   nominal flight
  Each day is a whole forecast wind profile; the gust is the 10 m forecast for the launch hour, used only for the rail exit AoA in gust.
```





PANDORA - AMA_CONFIG_15 - Inputs

```
[Case]
  Units               IN
  Analyze             STACK

[Nose]
  Shape               CONICAL
  Length              15
  PowerExponent       0.5
  TipRadius           0.1
  FlatTip             false

[Body]
  Diameter            2.91
  Length              52.29
  BoattailLength      0
  BoattailDiameter    3
  NozzleExitDiameter  2.5

[Fins]
  Count               4
  RootChord           9
  TipChord            5
  Span                3.65
  Sweep               5.5
  AftOffset           0
  Thickness           0.147
  Section             CHAMFER
  LEChamfer           1.019
  TEChamfer           0.7
  Cant                0

[FinCan]
  Enabled             true
  Diameter            3.15
  Length              10.05
  AftOffset           0
  FrontBevel          1.09

[Booster]
  Enabled             true
  Diameter            4.25
  Length              56.5
  Insertion           0
  TransitionLength    1
  NoseCone            false
  NoseShape           OGIVE
  NoseLength          8
  BoattailLength      0
  BoattailDiameter    0.01
  NozzleExitDiameter  3.5

[BoosterFins]
  Count               4
  RootChord           10
  TipChord            5
  Span                6
  Sweep               5
  AftOffset           0
  Thickness           0.187
  Section             CHAMFER
  LEChamfer           1
  TEChamfer           0.75
  Cant                0

[BoosterFinCan]
  Enabled             true
  Diameter            4.5
  Length              10.5
  AftOffset           0
  FrontBevel          0.5

[Protuberance1]
  Type                FAIRING
  Stage               SUSTAINER
  X                   17
  Count               2
  Length              1
  Width               0.4
  Height              0.4
  Standoff            0

[Protuberance2]
  Type                BUTTON
  Stage               BOOSTER
  X                   8
  Count               2
  Length              1
  Width               0.65
  Height              0.65
  Standoff            0

[Protuberance3]
  Type                LUG
  Stage               BOOSTER
  X                   50
  Count               2
  Length              1.5
  Width               0.65
  Height              0.65
  Standoff            0

[Flight]
  Altitude            4000
  ClipFinsToShock     false
  TurbulentBoundaryLayertrue

[Motors]
  Sustainer           N3364.eng
  Booster             O8220.eng

[Site]
  Weather             OPENMETEO
  Latitude            40.8707
  Longitude           -119.107
  Elevation           3701
  Date                2026-09-16
  Time                06:00
  Timezone            America/Los_Angeles
  WindSpeed           0
  WindDirection       0
  CylinderRadius      17

[Launch]
  RailLength          15
  Angle               89.045
  Heading             71
  Coast               14.939
  MaxTilt             10
  MinIgnitionAltitude 0
  MaxTime             1000
  MaxAoA              2
  AoAQ                0

[Recovery]
  ApogeeChuteDiameter 8
  ApogeeChuteCd       0.8
  MainChuteDiameter   28
  MainChuteCd         1.8
  MainChuteAltitude   2000

[Thermal]
  IncludeDescent      false

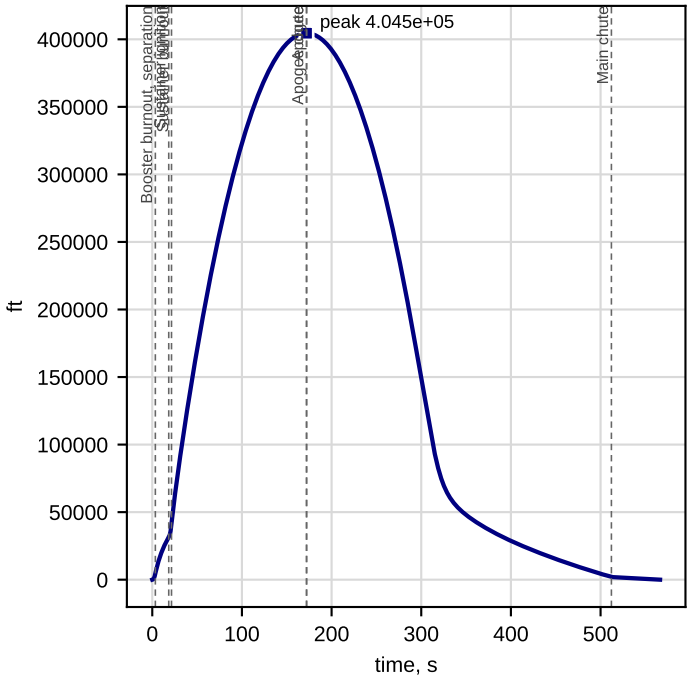
[DrySustainer]
  Mass                10.092
  CG                  41.9
  Inertia             4500
  RollInertia         32.021
  CgOffset            0.06
  AxisTilt            0.015
  MotorOffset         2.6

[DryStack]
  Mass                34.387
  CG                  85
  Inertia             30212.9
  RollInertia         194.192
  CgOffset            0.15
  AxisTilt            0
  MotorOffset         3.5

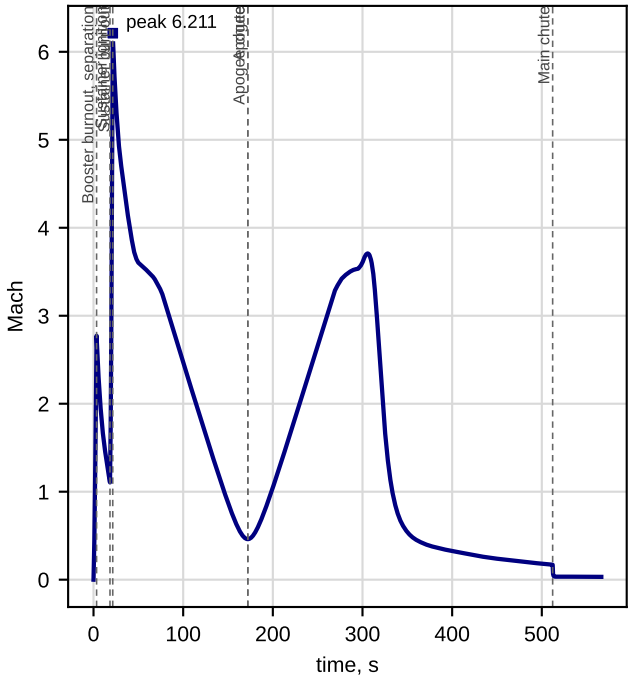
[MonteCarlo]
  Runs                0
  Seed                0
  Prediction          75
  AngleStd            0.5
  HeadingStd          180
  CoastStd            0.5
  BoosterImpulseStd   2
  SustainerImpulseStd 2
  BoosterMassStd      2
  SustainerMassStd    2
  CantStd             0.05
  SustainerFinAlignStd0.05
  BoosterFinAlignStd  0.05
  SustainerThrustOffsetStd0.05
  BoosterThrustOffsetStd0.1
  SustainerCgOffsetStd0.005
  BoosterCgOffsetStd  0.01
  SustainerTiltStd    0
  BoosterTiltStd      0
  WindSpeedStd        0
  WindDirectionStd    0
  WindDays            5
  MaxDayWind          10
  SkipDays
```

AMA_CONFIG_15 - Flight

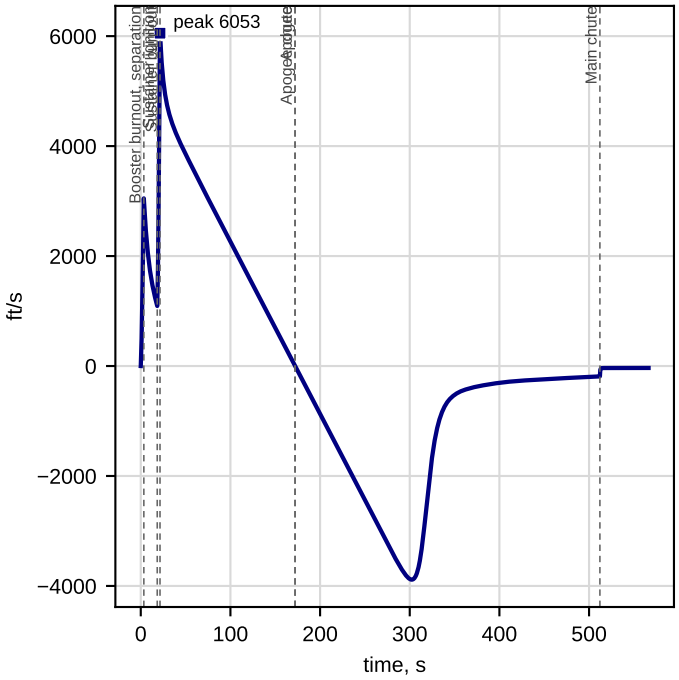
Altitude



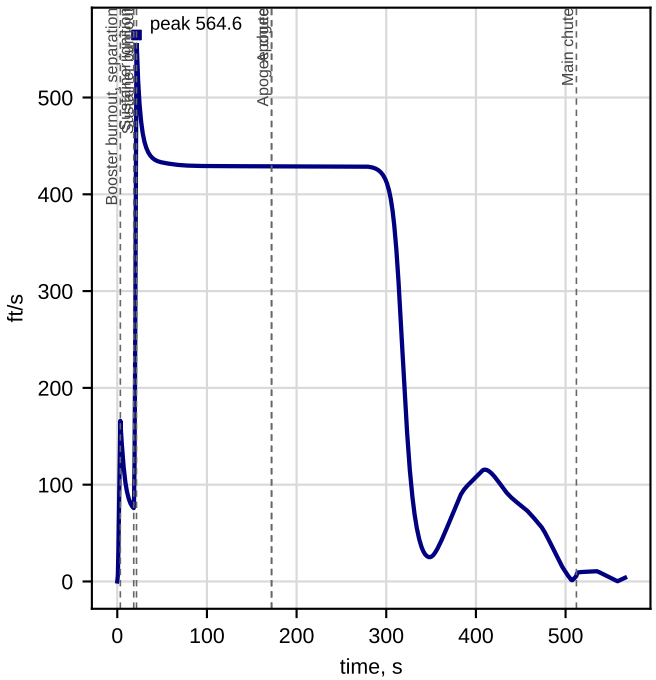
Mach number



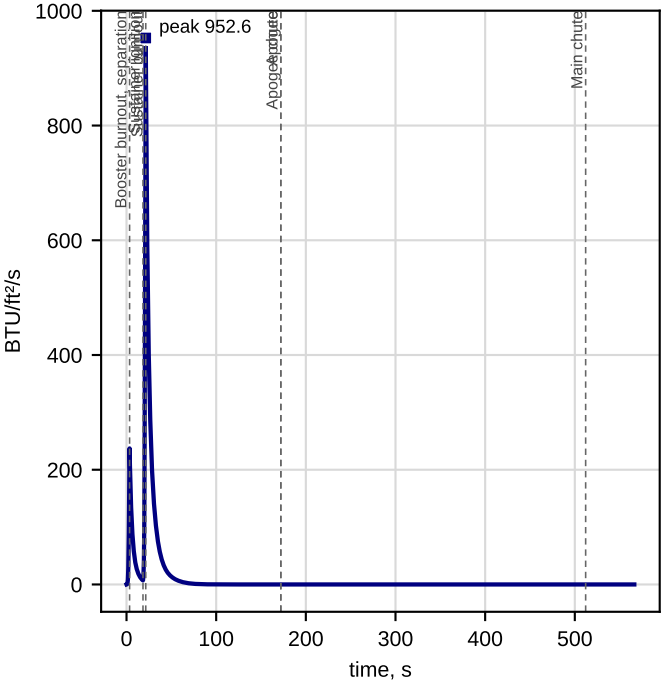
Vertical velocity



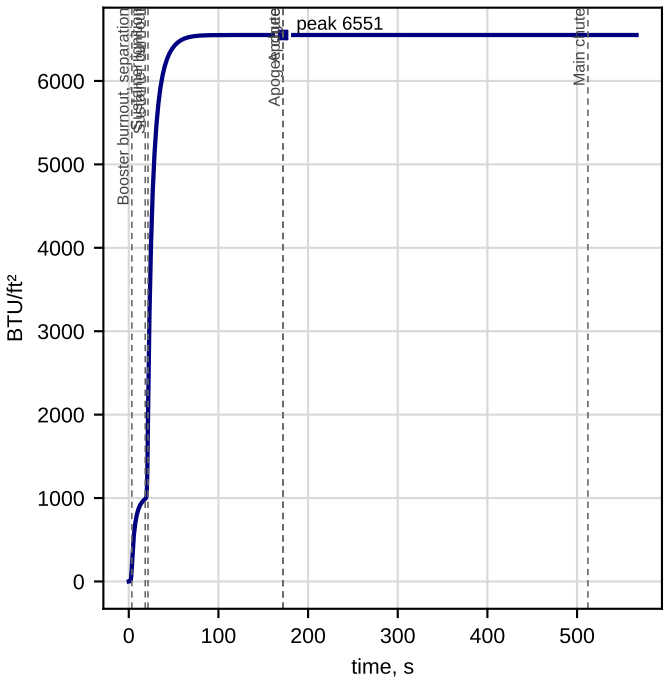
Horizontal velocity



Stagnation heat flux

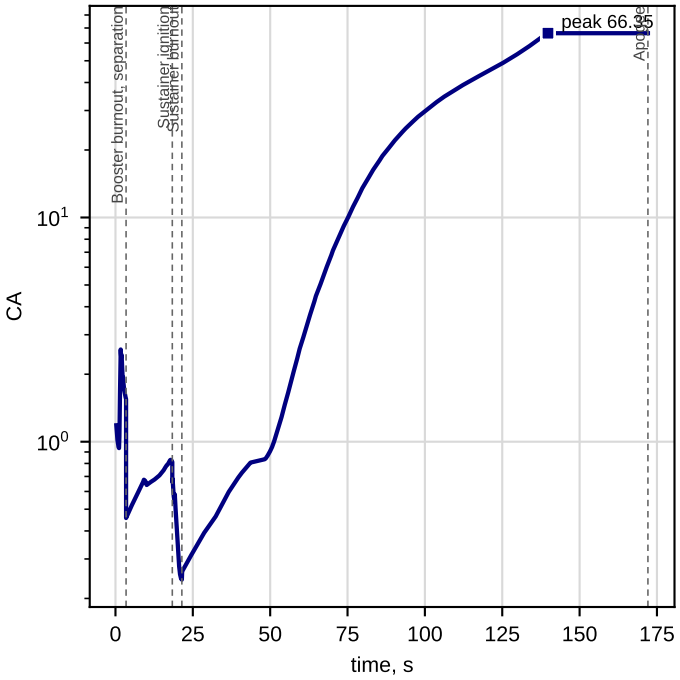


Integrated heat load

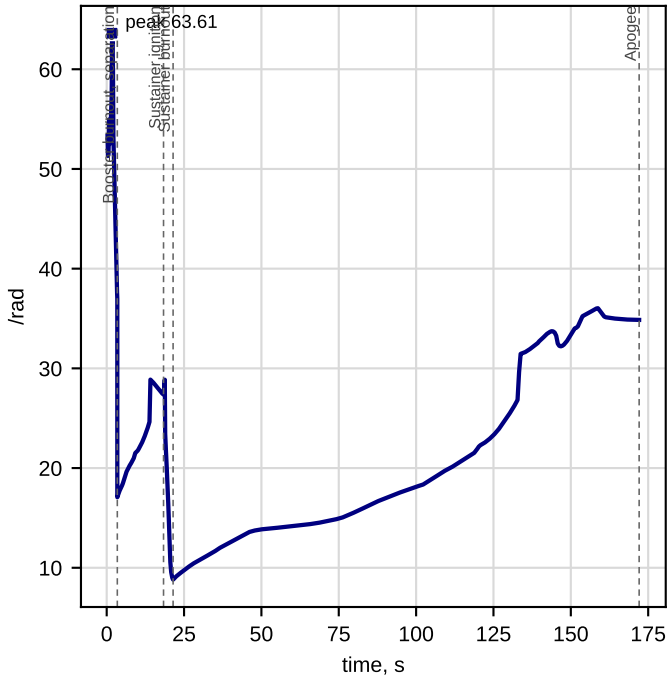


AMA_CONFIG_15 - DATCOM aerodynamics along the flight (launch to apogee, Mach 0.1 and up)

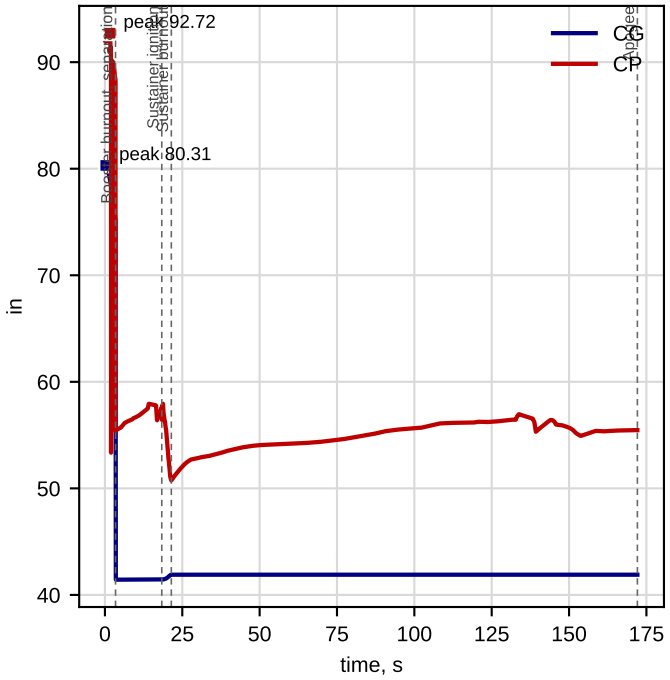
Axial force coefficient CA (log scale)



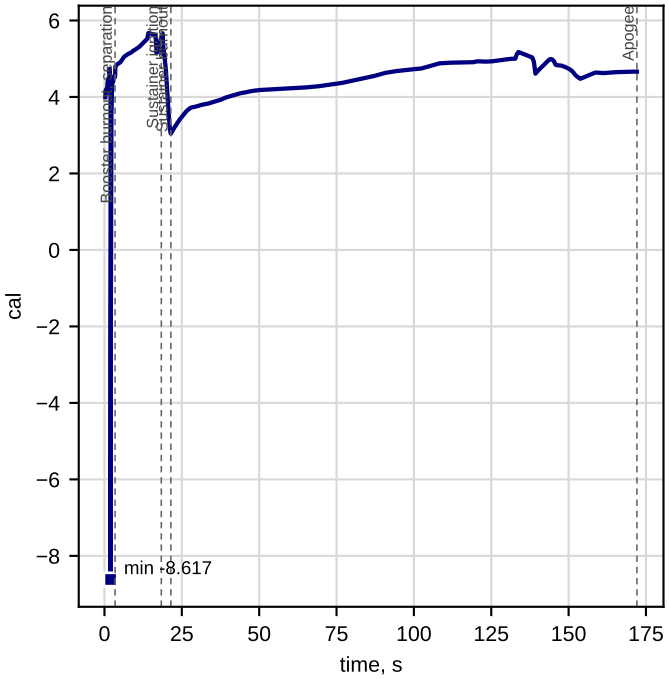
Normal force slope CNα



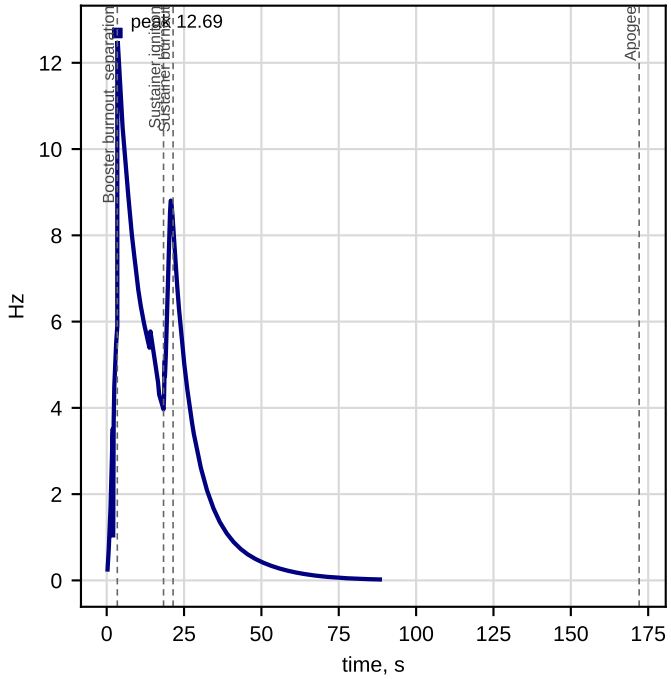
CG and centre of pressure



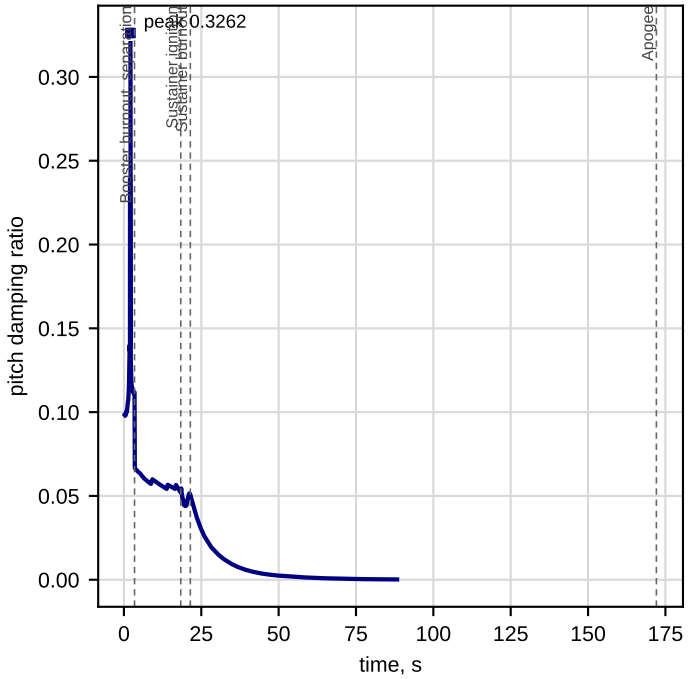
Static margin



Pitch natural frequency



Pitch damping ratio



AMA_CONFIG_15 - Ground track

